PRELIMINARY ENGINEERING RESEARCH PLAN

Adaptive, Compliance-Aware Parking Allocation for Shared Micromobility

A vision-assisted tabletop prototype and simulation for reducing scooter-corral overflow without imposing excessive walking or travel inconvenience on riders

Internal discussion draft only. This document was prepared with AI assistance to support early project selection and discussion. It is not intended for submission to ISEF, a regional science fair, or a school assignment. Before any submission, the student should independently verify the literature, redesign the work as needed, perform the project, maintain a contemporaneous notebook, and write the final research plan in the student's own words under the applicable fair's AI and authorship policies.

Proposed project type: Engineering design and testing

Proposed scope: Computer simulation plus a small tabletop computer-vision prototype; no street deployment

Working research question: Can a predictive, compliance-aware allocation algorithm that measures effective rather than nominal corral capacity reduce micromobility overflow while keeping rider inconvenience within a predefined limit?

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Important: Proposed procedures may need revision after review by the student's adult sponsor, school, and Scientific Review Committee. Approval must be obtained before experimentation whenever required by the current ISEF rules.

Rationale

Shared e-scooters and e-bikes provide short-distance and first/last-mile transportation, but dockless or partially docked systems also create a recurring public-space management problem. Vehicles are deposited and retrieved at unpredictable times, demand concentrates near transit, entertainment districts, campuses, and events, and riders do not always use designated parking. A scooter that is left outside a corral, placed diagonally, or knocked over can obstruct a sidewalk and disproportionately affect people using wheelchairs, canes, strollers, or other mobility aids.

Existing evidence shows that this is not solely a matter of providing a painted box. In New York City's East Bronx pilot, 24% of 13,970 inspected scooters were illegally or incorrectly parked, and 10% of all inspected scooters blocked the pedestrian path of travel (New York City Department of Transportation [NYC DOT], 2022). During unusual demand surges, fixed capacity can also fail: cities have reported corrals filling around major events, followed by sidewalk overflow and costly manual removal.

Research has established that dense, well-distributed parking infrastructure improves compliance. Meng et al. (2025) found a pronounced improvement around 20–30 corrals per square kilometer, while Sandoval et al. (2021) demonstrated data-driven methods for locating facilities and estimating required capacity. Other work predicts destinations and recommends stations with available space (Yan et al., 2024). These studies provide important foundations, but a practical gap remains for a student engineering investigation: nominal capacity may not equal usable capacity when scooters are misaligned or fallen, sensing is imperfect, several riders arrive together, and some riders reject inconvenient recommendations.

Scientifically, the project investigates a dynamic resource-allocation problem under uncertain demand, imperfect sensing, variable physical capacity, and incomplete compliance. The engineering importance lies in testing the entire feedback loop—sense, predict, allocate, communicate, and verify—rather than optimizing an initial layout only. Societally, a successful approach could reduce sidewalk obstruction and operator rebalancing while preserving the convenience that makes shared micromobility useful. The same methods may also apply to bicycle parking, event equipment storage, and other self-service spaces that deteriorate under repeated deposits and removals.

Research Question / Hypotheses / Engineering Goals / Expected Outcomes

Primary research question

Under normal and surge-demand conditions, can a vision-assisted, predictive, compliance-aware parking allocation algorithm reduce scooter-corral overflow and obstruction compared with static and reactive allocation methods while limiting riders' additional walking and travel distance?

Engineering goal

Design and validate a tabletop prototype and simulation that (1) detects scooter count, position, orientation, and fallen status; (2) converts those observations into an estimate of each corral's effective remaining capacity; (3) predicts short-horizon arrivals and departures; and (4) recommends a corral using overflow risk, distance, sensing uncertainty, and estimated compliance. If no acceptable permanent corral is available within the detour limit, the controller will activate a predefined temporary overflow area in the model.

Hypotheses and quantitative design criteria

1. **H1—Overflow:** The proposed controller will reduce overflow events per 100 arrivals by at least 25% relative to a reactive “nearest currently available corral” controller when evaluated on identical simulated traffic sequences.
2. **H2—Inconvenience:** The reduction in overflow will be achieved with mean additional walking distance no greater than 100 m and a 95th-percentile additional walking distance no greater than 200 m in the represented street network.
3. **H3—Robustness:** The proposed controller will retain a measurable overflow advantage when recommendation compliance falls to 70% and when simulated detection errors reach 10%.
4. **H4—Sensing:** The tabletop vision system will detect occupied scooter positions and classify acceptable versus disordered parking with an F1 score of at least 0.90 under the selected lighting and camera conditions.

Expected outcomes

It is expected that static allocation will perform adequately during balanced traffic but fail during concentrated arrivals. Reactive allocation should reduce some overflow but may send multiple approaching riders toward the same remaining space. The predictive controller is expected to reduce overflow by considering vehicles already en route and by distinguishing effective from nominal capacity. The likely tradeoff is a modest increase in walking distance. These are predictions, not results; the experiment may show that the added complexity does not provide a meaningful advantage or that performance is highly sensitive to compliance.

Materials

Material	Quantity / specification	Purpose
Laptop or desktop computer	1; capable of running Python and processing webcam video	Simulation, computer vision, control algorithm, storage, and analysis
USB webcam	1; 1080p preferred, 30 frames/s or greater	Overhead observation of the entire tabletop model
Camera support	1 tripod, rigid stand, or securely clamped overhead mount	Maintains a fixed camera position
Tabletop base	1 foam board or poster board, approximately 60 × 90 cm	Represents a small street network with multiple corrals
Corral and overflow-zone markings	Printed paper or removable tape; 5 permanent corrals and 1–2 temporary zones	Defines boundaries and known nominal capacities
Model scooter tokens	20–30 identical cardboard, wood, or 3D-printed tokens	Represents shared scooters; elongated shape permits orientation measurement
AprilTag or ArUco fiducial labels	One unique printed label per token plus calibration markers	Reliable identification, position, and angle estimation
Measurement and assembly supplies	Metric ruler, removable tape, paper, printer, blunt-tip scissors	Model construction and calibration
Software	Python 3; OpenCV; NumPy; pandas; SciPy; Matplotlib or equivalent	Vision, simulation, statistics, and graphing; open-source alternatives may be substituted
Optional indicators	1 ESP32 microcontroller and 5–10 low-voltage LEDs with resistors	Optional physical display of recommended or full corrals; not required for core testing

No chemicals, microorganisms, biological agents, vertebrate animals, controlled substances, firearms, radiation sources, or hazardous materials are proposed.

Procedures

1. Finalize scope and criteria before testing

1. Review the current ISEF rules with the adult sponsor and obtain all required approvals before experimentation.
2. Record the engineering requirements before seeing results: five permanent corrals, a maximum represented rider detour, effective-capacity detection, a temporary overflow option, and the quantitative success criteria stated above.
3. Document prior work on corral placement, parking compliance, recommendation systems, event geofencing, and computer-vision parking validation. Revise the novelty claim so it describes only the student's tested contribution.

2. Construct and calibrate the tabletop model

1. Draw or print a simplified network with five corrals. Assign each corral a nominal capacity between 4 and 10 tokens and a distance matrix representing approximately 50–250 m between destinations and corrals.
2. Mark one or two temporary overflow areas that are inactive unless the proposed controller determines that no permanent corral can satisfy both capacity and detour constraints.
3. Attach a unique fiducial label to each elongated scooter token. Mount the webcam so all corrals and calibration points remain visible. Lock focus and exposure if the camera permits.
4. Convert image coordinates to tabletop coordinates with a planar perspective transform based on four or more fixed calibration markers. Record calibration error using known ruler distances.

3. Develop effective-capacity sensing

1. Detect each token's identity, center point, and orientation angle. Classify a token as inside, partially outside, or outside a corral polygon.
2. Define "orderly" before testing: for example, the token is fully inside the boundary, upright in the model, and within 30° of the designated alignment axis. Record alternative thresholds in a sensitivity analysis rather than changing them after observing outcomes.
3. Estimate effective capacity from unoccupied usable footprints, accounting for diagonal, overlapping, fallen, or boundary-crossing tokens. Preserve the raw image detections so the calculation can be audited.
4. Create a ground-truth image set containing empty, partially occupied, full, disordered, and overflowed arrangements under at least three lighting levels. Hand-label the true token states before running the vision classifier.

4. Build the traffic simulation

1. Generate synthetic arrivals and departures for a 60-minute modeled period. Use non-homogeneous Poisson arrivals or another documented stochastic process so the rate may change over time. Use a fixed, documented distribution for scooter dwell time before retrieval.
2. Create three traffic profiles:
    - **Normal:** relatively balanced arrivals and departures;
    - **Commuter peak:** gradual concentration near one or two destinations; and
    - **Event release:** a sharp, synchronized arrival surge near one destination.

3. Model imperfect compliance at nominal rates of 100%, 85%, 70%, and 50%. A noncompliant simulated rider will use the nearest convenient location according to a predefined rule. These values are sensitivity-test assumptions and will not be presented as measurements of real rider behavior.
4. Inject 0%, 5%, and 10% missed or false occupancy detections. Separately vary the proportion of disordered placements so sensing error and physical disorder are not confused.
5. Save all random seeds. Every controller must receive the same arrival, departure, destination, compliance, and disorder sequence within a matched trial.

5. Implement four allocation conditions

Condition	Decision rule	Role in experiment
A. Free parking	Riders use the nearest convenient location; no allocation guidance	Unmanaged baseline
B. Static corrals	Riders are directed to the nearest permanent corral without live capacity information	Infrastructure-only baseline
C. Reactive allocation	Select the nearest corral that appears to have space at the instant of recommendation	Live-data baseline
D. Proposed controller	Predict short-horizon occupancy, use effective capacity and uncertainty, estimate acceptance, enforce a detour cap, and activate temporary overflow if necessary	Experimental design

The proposed controller will calculate a documented score such as:

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Score(j) = α·P(overflow at arrival) + β·added walking distance + γ·obstruction risk +
δ·sensing uncertainty + ε·predicted noncompliance cost
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The coefficients will be selected using a training subset or an explicitly defined engineering tradeoff and then frozen before final evaluation. Test trials will not be used to tune the controller. Corral choices exceeding the predefined detour limit will be rejected.

6. Run simulation trials

1. Run 100 matched replications for every combination of four controllers, three traffic profiles, four compliance levels, and three sensing-error levels. This produces 14,400 controller runs, but only 3,600 independently generated traffic sequences because each sequence is reused across the four controllers.
2. Log every event: time, destination, recommendation, predicted capacity, actual capacity, compliance decision, final parking location, overflow status, obstruction status, added distance, and overflow-zone activation.
3. Keep a versioned copy of code, parameters, seeds, and unprocessed output. Record software changes in the project notebook.

7. Conduct physical validation trials

1. Select ten previously unseen random sequences for each of the three traffic profiles. Use a fixed 70% or 85% compliance level selected before physical testing.
2. For each sequence, run all four controllers. The student will move tokens at the generated event times exactly as specified by the saved script. No other person will act as a test subject.

3. Randomize controller order to reduce learning or handling-order effects. Reset the board from a saved initial-state photograph before each trial.
4. Compare camera-derived states with manually verified ground truth after the trial. Do not correct outcomes manually during a run unless the correction rule was defined beforehand.
5. Repeat trials affected by an unrelated equipment failure only under a written repeat rule, preserving the failed run and reason for exclusion.

8. Data sources and reproducibility

Primary performance data will be generated by the student's simulation and tabletop prototype. Published city statistics will establish realistic ranges and context but will not be represented as the student's experimental data. If a public trip dataset is later used for calibration, its exact URL, retrieval date, license, fields, exclusions, and preprocessing steps will be recorded. Only anonymized or aggregated mobility data will be used. Source code, parameter files, random seeds, and analysis scripts will be retained so each result can be reproduced.

Risk Assessment

Potential risk	Risk level	Precaution
Scissors or model-construction tools	Low	Use blunt-tip scissors when possible; cut away from the body; adult supervision for any sharper tool.
Falling camera stand or trailing cable	Low	Secure or clamp the stand, keep it away from table edges, and tape down cables.
Low-voltage electronics, if optional LEDs are used	Low	Use USB/low-voltage power only, disconnect before rewiring, inspect for shorts, and use correct resistors.
Eye strain or repetitive computer work	Low	Use regular breaks and an ergonomically suitable workstation.
Camera privacy	Low	Point the camera only at the tabletop, crop the field of view, process locally, and do not record faces or voices.
Misinterpretation as a proven public-safety system	Project limitation	Label the prototype as a scale model; do not test in streets or direct real riders; report limits and failure cases.

No human participants are included in the proposed experiment. The student will manipulate tokens according to computer-generated schedules; the student is the investigator, not a study subject. No survey, interview, observation of identifiable riders, or behavioral intervention will be conducted. If any human-participant component is later proposed—for example, measuring willingness to walk to another corral—it must be designed separately and receive Institutional Review Board approval before recruitment or data collection. No real-road or real-scooter test is proposed.

Data Analysis

Primary outcome

Overflow events per 100 arrivals, defined as an arrival for which the final placement exceeds the usable corral boundary or capacity. The exact geometric rule will be fixed before final testing.

Secondary outcomes

- obstructing or boundary-crossing placements per 100 arrivals;
- mean and 95th-percentile additional walking distance and estimated time;
- recommendation acceptance and noncompliant-parking rates;
- temporary overflow activations and modeled rebalancing interventions;
- corral utilization and utilization imbalance;
- forecast error for occupancy at arrival; and
- vision precision, recall, F1 score, position error, angle error, and capacity-estimation error.

Comparisons

1. Construct summary tables by controller, traffic profile, compliance level, and sensing-error level. Report means, medians, standard deviations or interquartile ranges, and 95% confidence intervals as appropriate.
2. Because controllers receive identical random sequences, calculate paired differences between the proposed controller and each baseline. Use bootstrap 95% confidence intervals for the paired mean or median difference.
3. Examine controller, demand, compliance, and sensing-error effects using a factorial model. A count model may be used for overflow counts; if simpler parametric tests are used, check their assumptions. If assumptions are not met, use an appropriate nonparametric paired alternative and report effect sizes, not only p-values.
4. Plot overflow against added distance to create a Pareto tradeoff graph. A controller is more useful if it lowers overflow without an unacceptable increase in inconvenience.
5. Produce line graphs for performance versus compliance and sensing error, boxplots for matched trial outcomes, heat maps of corral occupancy over time, and confusion matrices for vision classification.
6. Compare results with the prespecified criteria. A statistically detectable reduction that violates the detour requirement will not count as engineering success.

Simulation and physical-prototype results will be reported separately before being compared. Disagreements will be treated as evidence of model limitations rather than silently pooled.

Bibliography

APA-style working bibliography. Publication details and live links should be independently verified before use in a final student document.

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Additional Project Information

- **Human participants:** None in the proposed design.
- **Vertebrate animals:** None.
- **Potentially hazardous biological agents:** None.
- **Hazardous chemicals, activities, or devices:** None beyond ordinary low-risk craft work, a secured camera stand, and optional low-voltage electronics.
- **Continuation project:** Assumed to be a new project. If it continues earlier student work, Form 7 requirements and prior-year materials must be addressed.
- **Regulated research institution:** None anticipated. Any later work performed in a university, company, city agency, or other regulated setting must be disclosed and reviewed.

- **Public or third-party data:** Not required for the primary experiment. Any later dataset must be documented with source, permission/license, privacy characteristics, and preprocessing.
- **Real-world deployment:** Outside the proposed scope. The prototype will not direct public riders, modify public property, or operate on a street.

> **Approval and authorship note:** This draft does not determine which forms or approvals are required. The student and adult sponsor should use the current year's official ISEF rules and local fair requirements. Any later addition of surveys, interviews, observation of identifiable people, behavioral testing, public-road activity, or private operator data may change the review requirements and must be approved before that work begins.